

Derivatives (Univariate): A Rigorous Introduction

Math Foundations Map — Node 12

First, an example

Before any definitions, let us compute a derivative by hand. Take $f(x) = x^2$ and ask: what is $f'(2)$? Working straight from the limit-of-the-difference-quotient idea,

$$f'(2) = \lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{h} = \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4}{h} = \lim_{h \rightarrow 0} (4 + h) = 4.$$

Read aloud: “f-prime of two equals the limit as h goes to zero of two plus h, squared, minus four, over h; expanding gives four plus four h plus h-squared minus four, over h; the constants cancel and dividing by h leaves four plus h, whose limit as h tends to zero is four.” So the slope of $y = x^2$ at the point $(2, 4)$ is 4. The rest of the lesson formalises what “slope at a point” means and what we are allowed to do with it.

1 Motivation

Given a function $f : I \rightarrow \mathbb{R}$ on an open interval I and a point $c \in I$, we wish to assign a number $f'(c)$ that captures the *instantaneous rate of change* of f at c . Geometrically, this number is the slope of the line that best approximates the graph of f in an arbitrarily small neighbourhood of $(c, f(c))$. Algebraically, it is the limit of the slopes of secant lines through $(c, f(c))$ as the second point collapses onto $(c, f(c))$.

2 Definitions

Definition 1 (Derivative at a point). Let $f : I \rightarrow \mathbb{R}$ where $I \subseteq \mathbb{R}$ is an open interval, and let $c \in I$. We say f is **differentiable at c** if the limit

$$f'(c) := \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

Read aloud: “f-prime of c is defined as the limit as h tends to zero of f of c plus h minus f of c, all over h.” exists in $\mathbb{R}$. The value $f'(c)$ is called the **derivative** of f at c . An equivalent formulation, obtained by substituting $x = c + h$, is

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

Read aloud: “f-prime of c equals the limit as x approaches c of f of x minus f of c, all over x minus c.”

Definition 2 (Differentiability on an interval). f is **differentiable on I** if $f'(c)$ exists for every $c \in I$. In this case the assignment $c \mapsto f'(c)$ defines a function $f' : I \rightarrow \mathbb{R}$, called the **derivative function**.

Definition 3 (Local linear approximation). f is differentiable at c with derivative m if and only if there exists a function ε defined on a neighbourhood of 0 with $\lim_{h \rightarrow 0} \varepsilon(h) = 0$ and

$$f(c+h) = f(c) + m \cdot h + h \cdot \varepsilon(h).$$

Read aloud: “ f of c plus h equals f of c , plus m times h , plus h times epsilon of h .”

3 Differentiability implies continuity

The first nontrivial fact about differentiation is that it strengthens continuity. We isolate this as the main theorem of the lesson.

Theorem 4 (Differentiable implies continuous). *If $f : I \rightarrow \mathbb{R}$ is differentiable at $c \in I$, then f is continuous at c .*

Proof. We must show $\lim_{x \rightarrow c} f(x) = f(c)$, equivalently $\lim_{x \rightarrow c} (f(x) - f(c)) = 0$. For $x \in I$ with $x \neq c$, we use the algebraic identity

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c). \quad (1)$$

Read aloud: “ f of x minus f of c equals the difference quotient, f of x minus f of c over x minus c , times the quantity x minus c .” By the definition of $f'(c)$,

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = f'(c) \in \mathbb{R}.$$

Read aloud: “the limit as x approaches c of f of x minus f of c , all over x minus c , equals f -prime of c , which lies in the real numbers.” Also, trivially,

$$\lim_{x \rightarrow c} (x - c) = 0.$$

Read aloud: “the limit as x approaches c of x minus c equals zero.” The product rule for limits (which holds whenever both factors have finite limits) applied to (1) yields

$$\lim_{x \rightarrow c} (f(x) - f(c)) = \left(\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right) \cdot \left(\lim_{x \rightarrow c} (x - c) \right) = f'(c) \cdot 0 = 0.$$

Read aloud: “the limit as x approaches c of f of x minus f of c equals the limit of the difference quotient times the limit of x minus c , which is f -prime of c times zero, which is zero.” Therefore $\lim_{x \rightarrow c} f(x) = f(c)$, i.e. f is continuous at c . $\square$

Remark 5. The converse is false. The function $f(x) = |x|$ is continuous at 0 but the two one-sided limits of the difference quotient are $+1$ and -1 , so $f'(0)$ does not exist.

4 Differentiation rules

Proposition 6 (Algebraic rules). *Let f, g be differentiable at c , and let $\alpha \in \mathbb{R}$. Then:*

1. $(\alpha f)'(c) = \alpha f'(c)$;
2. $(f + g)'(c) = f'(c) + g'(c)$;

3. $(fg)'(c) = f'(c)g(c) + f(c)g'(c);$
4. if $g(c) \neq 0$, $(f/g)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}.$

Proposition 7 (Chain rule). *If g is differentiable at c and f is differentiable at $g(c)$, then $f \circ g$ is differentiable at c and $(f \circ g)'(c) = f'(g(c)) \cdot g'(c)$.*

5 The Mean Value Theorem

Theorem 8 (Rolle). *If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $\xi \in (a, b)$ with $f'(\xi) = 0$.*

Theorem 9 (Mean Value Theorem). *If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $\xi \in (a, b)$ with*

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

Read aloud: “ f -prime of ξ equals f of b minus f of a , all over b minus a .”

The MVT is the bridge between local information (f') and global behaviour ($f(b) - f(a)$). Consequences include: monotonicity criteria, L^∞ bounds on increments, Taylor’s theorem with Lagrange remainder, and the uniqueness of antiderivatives up to additive constants.

6 Example: $\frac{d}{dx}x^2$ from first principles

Example 10. For $f(x) = x^2$ and $c \in \mathbb{R}$,

$$\frac{f(c+h) - f(c)}{h} = \frac{(c+h)^2 - c^2}{h} = \frac{2ch + h^2}{h} = 2c + h \xrightarrow{h \rightarrow 0} 2c.$$

Read aloud: “ f of c plus h , minus f of c , all over h , equals the quantity c plus h squared minus c squared, over h ; equals two c h plus h squared, over h ; equals two c plus h , which tends to two c as h goes to zero.” Hence $f'(c) = 2c$, i.e. $\frac{d}{dx}x^2 = 2x$.